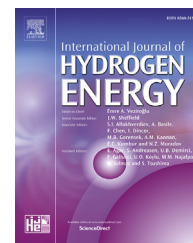


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Design of a large-scale metal hydride based hydrogen storage reactor: Simulation and heat transfer optimization

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ABSTRACT

An optimized design for a 210 kg alloy, Ti–Mn alloy based hydrogen storage system for stationary application is presented. A majority of the studies on metal hydride hydrogen systems reported in literature are based on system scale less than 10 kg, leaving questions on the design and performance of large-scale systems unanswered. On the basis of sensitivity to various design and operating parameters such as thermal conductivity, porosity, heat transfer coefficient etc., a comprehensive design methodology is suggested. Following a series of performance analyses, a multi-tubular shell and tube type storage system is selected for the present application which completes the absorption process in 900 s and the desorption process in 2000 s at a system gravimetric capacity of 0.7% which is a vast improvement over similar studies. The study also indicates that after fifty percent reaction completion, heat transfer ceases to be the major controlling factor in the reaction. This could help prevent over-designing systems on the basis of heat transfer, and ensure optimum system weight.

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Introduction

Metal hydrides are a favorable hydrogen storage alternative offering safe and efficient storage. Hydrogen reacting with a metal or metal alloy, forms a metal hydride, and thus gets stored in solid state. Of the various metal hydrides, AB₂ type Ti-based metal hydrides have a promising potential because of their comparatively high hydrogen storage capacity, fuel cell friendly operating temperatures and pressures, and good reversibility [1]. However, the storage of hydrogen in metal hydrides is primarily limited by heat transfer associated with the reaction. While the absorption reaction is accompanied by release of heat which needs to be removed from the system;

for desorption to occur heat needs to be supplied to the system. This reaction heat has a great bearing upon the system performance as it controls the MH bed temperature which in turn controls the equilibrium pressure. The rate of absorption/desorption of hydrogen is a function of the equilibrium pressure. If the equilibrium pressure is higher than the supply pressure of hydrogen the absorption reaction will cease. Similarly if the equilibrium pressure is lower than the discharge pressure of hydrogen desorption reaction will be stalled. Therefore, heat transfer to/from the MH bed is an important factor for ensuring efficient hydrogen storage.

Multiple studies have addressed the issue of heat transfer in metal hydride based hydrogen storage. Studies have suggested various designs for hydrogen storage systems to

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Nomenclature

c	Fraction of hydrogen absorbed
M	Rate of absorption/desorption, kg/m ³ s
P	Pressure, bar
T	Temperature, K
u	Velocity, m/s
R	Universal gas constant, J/mol/K
ΔS	Entropy of reaction, J/mol/K
ΔH	Enthalpy of reaction, J/mol
ε	Porosity
h	Heat transfer coefficient, W/m ² K
C _p	Specific heat capacity, J/kg/K
K	Thermal conductivity, W/m/K
Q	Heat source/sink, W/m ³

Subscripts

eq	equilibrium
f	final
e	effective
m	metal
g	gas
s	source/sink
a	absorption
d	desorption

Abbreviations

MH	Metal hydride
ENG	Expanded natural graphite
HTF	Heat transfer fluid
PCM	Phase changing materials
HPMH	High pressure metal hydride

facilitate effective heat transfer. These designs are as diverse as the heat transfer enhancements themselves and include addition of fins, cooling tubes, cooling coils etc. Attempts have been made to improve thermal conductivity of the MH bed by addition of foam and ENG (expanded natural graphite) [2–10], improve heat transfer area by addition of fins and cooling tubes [11–19]. More recently, combinations of these techniques have been employed, yielding favorable results. Garrison et al. studied the effect of a concentric cooling tubes equipped with a) transverse fins b) longitudinal fins on a sodium alanate system [20]. The results pointed to a configuration involving multiple cooling tubes at small distances as the most viable. Ferekh et al. compared the performance of metal fins versus metal foam and found that metal foam based design was superior to the fin based design [10]. Helical coil based heat exchangers have also been shown to improve system performances by reducing charging/discharging times significantly [21–24]. A comparative study involving finned cooling tube reactor, helical coil reactor and shell-tube type reactor was performed by Raju et al. Working with sodium alanate, for a filling time of 10.5 min they found that the helical coiled heat exchanger gave better volumetric and gravimetric densities over the other designs [25]. Maz-zucco et al. studied three different heat exchanger configurations: a) metal hydrides with embedded cooling tubes b) metal hydrides tubes surrounded by heat transfer fluid in

annular space and c) metal hydride tubes in a heat transfer fluid shell [26]. Their results indicated that the last configuration was of most significant interest for storage applications.

Clearly, there is no dearth of studies on design of metal hydride hydrogen storage systems but the studies on large scale systems are far less than those on small-scale systems [27]. Further investigation of large-scale practical systems is important keeping in view the issues that are unique to these systems such as creation of hot spots (pockets of extremely high temperature where heat transfer enhancements may not be able to reach), parasitic system weight, volumetric efficiency etc. More importantly, the design methodology for large-scale systems is also unclear. While every heat transfer enhancement results in improved system performance, it also leads to reduced gravimetric and volumetric capacities. Therefore, the question of which heat transfer enhancements should be preferred is very significant. In this study an attempt is made to present a unified design policy which can serve as a road-map for the design of such systems. Using a 100 kWh Ti–Mn based (Hydralloy C5) hydrogen storage system as an example, the most important parameters that affect the system performance are studied and a procedure outlined to serve as a tool for the design of future systems. The gravimetric capacities of the final designs are compared to ensure that there is a healthy trade-off between system performance and weight.

Design objective

The objective of this exercise is to design an optimized Ti based, 100 kWh metal hydride hydrogen storage system for stationary applications and to outline a design methodology for the same. The considerations governing the design process have been outlined in Table 1. To begin with, the volume required to store the metal alloy powder was evaluated on the basis of the mass of alloy required to store the hydrogen and the density of the alloy (5500 kg/m³) [1,28]. Assuming a porosity of 0.5, the volume was determined to be 0.09168 m³, which includes a twenty per cent expansion volume, to allow for the expansion of the alloy. A cylindrical design was chosen to ensure sufficient heat transfer area, ease in handling and portability. To keep the design within a conformable limit for convenient storage and transportation, the initial value of height of the reactor was limited to 1 m and the corresponding radius was evaluated. The closest standard size pipe dimension was then taken and the corresponding height was calculated. The thickness of the pipe was found to be suitable for the pressure range under consideration on the basis of the ASME code for pressure vessels. In this manner, the radius and height of the cylindrical reactor were found to be 0.1524 m and 1.047 m respectively.

Mathematical modeling

The reactor of the said dimensions was first simulated under free convection conditions and its performance was analyzed. The system was modeled as a 2-d axisymmetric domain. To

Table 1 – Design constraints and variables.

Variable	Constraint
Reactor Volume	H ₂ required to produce X kW power >> Alloy M required to store H ₂ >> System volume required to store MH
Reactor diameter, D	Standard pipe dimensions available
Reactor height, L	Conformability in terms of storage space availability etc.
Reactor thickness, t	Allowable stress for the tank

model the MH hydrogen storage system, mass, momentum and energy conservation equations are used.

The following assumptions have been used for modeling the system:

- 1) The solid and gas are at the same temperature (local thermal equilibrium).
- 2) Gas phase: hydrogen behaves as an ideal gas.
- 3) Solid phase: MH is isotropic and has uniform porosity.
- 4) The pressure inside the tank is uniform throughout.

The simulation domain consists of the metal hydride bed and the expansion volume. The reactor is exposed to air at a temperature T_f . The alloy considered for the model is AB₂ type, Ti–Mn based Hydralloy C5, chosen for its good kinetics and reversibility, and fuel cell friendly operating temperature and pressure. The model was developed in COMSOL 5.2. The details of governing equations and boundary conditions are as follows:

Continuity

The continuity equation for the expansion volume part which is the gas filled void space, and the metal hydride bed is represented by Eq. (4). Here ρ_g is the density of hydrogen gas and ρ_m is the density of the metal hydride. The density of the metal hydride depends upon M which is the rate of absorption/desorption of hydrogen per unit volume. Note that the M will be negative in Eq. (5) and positive in Eq. (6) for absorption and vice versa for desorption.

$$\text{Expansion Volume: } \frac{\partial \rho_g}{\partial t} + \nabla \cdot (\rho_g \cdot \mathbf{u}) = 0 \quad (4)$$

$$\text{MH bed: } \varepsilon \frac{\partial \rho_g}{\partial t} + \nabla \cdot (\rho_g \cdot \mathbf{u}) = -M \quad (5)$$

$$(1 - \varepsilon) \frac{\partial \rho_m}{\partial t} = M \quad (6)$$

Conservation of momentum

The conservation of momentum is described by the Navier-Stokes equation for the expansion volume, and the Brinkman equation for the metal hydride bed which reduces to Darcy law if viscous effects and gas velocity are low. Here μ_g , ε , and κ are the viscosity of the gas, the porosity and the permeability of the bed respectively; F is the body force. The variable $\mathbf{u}$ represents the velocity of hydrogen gas within the pores and the expansion volume domain.

$$\begin{aligned} \text{Expansion Volume: } & \rho_g \frac{\partial \mathbf{u}}{\partial t} + \rho_g (\mathbf{u} \cdot \nabla) \mathbf{u} \\ & = -\nabla p + \nabla \cdot \left[\mu_g (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} \mu_g (\nabla \cdot \mathbf{u}) \right] + F \end{aligned} \quad (7)$$

$$\begin{aligned} \text{MH bed: } & \frac{\rho}{\varepsilon} \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \frac{\mathbf{u}}{\varepsilon} \right] \\ & = -\nabla p + \nabla \cdot \left[\frac{\mu_g}{\varepsilon} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} \frac{\mu_g}{\varepsilon} (\nabla \cdot \mathbf{u}) \right] - \left(\frac{\mu_g}{\kappa} + \frac{M}{\varepsilon^2} \right) \mathbf{u} + F \end{aligned} \quad (8)$$

Conservation of energy

Assuming thermal equilibrium between gas and solid, T represents temperature within the system. The variables C_{pg} , k_g and k_m are specific heat of the gas, thermal conductivity of the gas and thermal conductivity of the porous bed. The effective thermal conductivity and the effective thermal mass of the porous bed and the gas is represented by k_e and $(\rho C_p)_e$.

$$\text{Expansion Volume: } \rho_g C_{pg} \left[\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T \right] = \nabla \cdot (k_g \nabla T) \quad (9)$$

$$\text{MH bed: } (\rho C_p)_e \frac{\partial T}{\partial t} + \rho_g C_{pg} (\mathbf{u} \cdot \nabla) T = \nabla \cdot (k_e \nabla T) + Q_s \quad (10)$$

$$\text{where, } (\rho C_p)_e = \varepsilon \rho_g C_{pg} + (1 - \varepsilon) \rho_m C_{pm} \quad (11a)$$

$$k_e = \varepsilon k_g + (1 - \varepsilon) k_m \quad (11b)$$

The reaction kinetics that determine the amount of hydrogen absorbed/desorbed are expressed as:

$$\text{Absorption: } M = C_a \exp\left(\frac{-E_a}{R_g T}\right) \ln\left(\frac{p}{p_{eqa}}\right) (\rho_{ss} - \rho_m) \quad (12)$$

$$\text{Desorption: } M = C_d \exp\left(\frac{-E_d}{R_g T}\right) \left(\frac{p - p_{eqd}}{p_{eqd}}\right) (\rho_m - \rho_e) \quad (13)$$

$$\begin{aligned} \text{where, } \frac{p_{eq}}{p_{ref}} &= \left(\exp\left(\frac{S}{R} - \left(\frac{H}{R T}\right)\right) + 0.51 \tan(\pi^*(c - 0.5)) \right. \\ & \quad \left. + 0.15 \right) \end{aligned} \quad (14)$$

The equilibrium pressures are predicted with the help of van't Hoff equation as stated in Eq. (14). Please note that 'c' is the fraction of hydrogen absorbed/desorbed.

Initial and boundary conditions

The evolution of temperature, hydrogen absorption/desorption in the metal bed is calculated with respect to certain initial and boundary conditions. Presently, the conditions for absorption only will be discussed. For absorption the initial temperature of the tank is considered to be 298 K and the pressure is 25 bar which is well above the equilibrium pressure of the alloy under consideration. For the tank center line, the axisymmetric boundary condition is set. Along the tank wall, a heat flux condition is set with constant heat transfer coefficient. A no-slip boundary condition is set along the tank

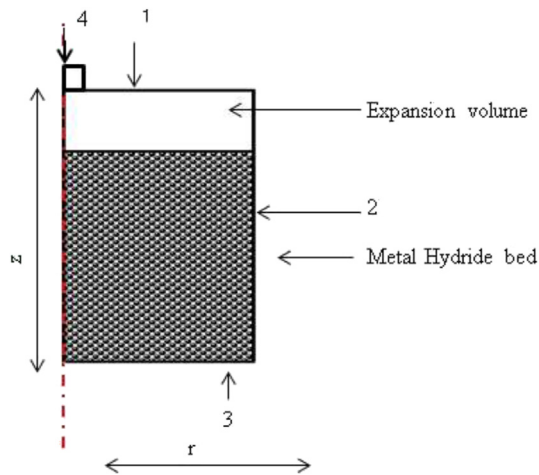


Fig. 1 – Boundary definition of the model MH tank.

walls except for the inlet through which hydrogen is allowed to enter. Continuity of temperature, density, velocity and other properties is established at the boundaries between the MH bed and the expansion volume. The boundary and initial conditions may be summarized as follows:

$$T(r, z, 0) = 298 \text{ K}; p(r, z, 0) = 25 \text{ bar}; \mathbf{u}(r, z, 0) = 0;$$

$$\rho m(r, z, 0) = 5500 \text{ kg/m}^3; \rho_g(r, z, 0) = \frac{p_g \cdot R_s}{T}$$

$$\text{At the center line, } u_r = 0; \frac{\partial u_z}{\partial z} = 0; \frac{\partial p_g}{\partial r} = 0$$

As can be seen from Fig. 1, boundaries 1 and 3 are assumed to be thermally insulated, while 2 is subjected to a heat flux through the cooling/heating fluid. Boundary 4 is exposed to the inlet pressure.

$$\text{At 1 and 3: } -k \frac{\partial T}{\partial n} = 0; \mathbf{u} = 0$$

$$\text{At 2: } -k \frac{\partial T}{\partial n} = h(T - T_f)$$

$$\text{At 4: } \mathbf{n} \cdot \nabla T = 0; p = p_{\text{inlet/outlet}}$$

Model validation

The mathematical model used is well established in literature [3,13,29–31]. However, it has also been validated in the present study as shown in Fig. 2. The results are compared to Laurencelle and Goyette [3]. The grid independence and time independence are shown in Figs. 3 and 4 respectively.

Results and discussions

The baseline mathematical model discussed in section Conservation of momentum is based upon the above governing equations and developed in COMSOL 5.2. The basic cylindrical reactor of radius 15.24 cm and height 1.256 m was simulated operating at a pressure of 25 bar and a temperature of 298 K under free convection, exposed to air at 298 K. Ordinarily, in the design of any heat exchanger, if the heat load is known, starting with an initial value of heat transfer coefficient, the area of the reactor can be found and further calculations can be made. But that approach cannot be taken here because the surface area requirement in that case is impossibly large. Therefore, the reverse procedure was followed: the performance of the system was studied under free convection conditions, the key parameters influencing system behavior were identified and the design was further improved. A parametric study was thus undertaken and the sensitivity of

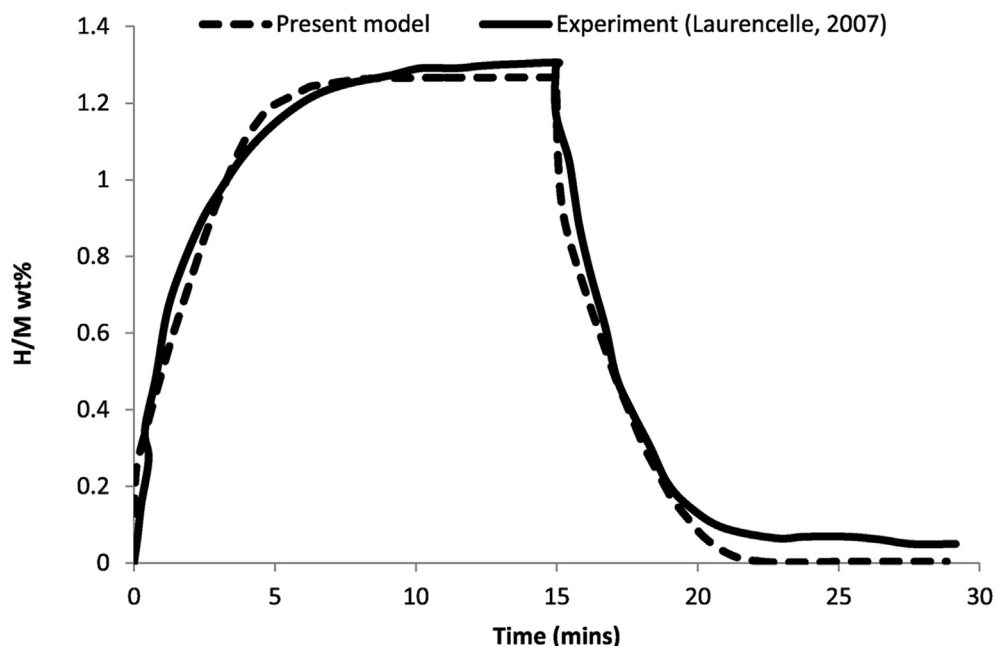


Fig. 2 – Comparison of H/M weight per cent from the present model and experimental data.

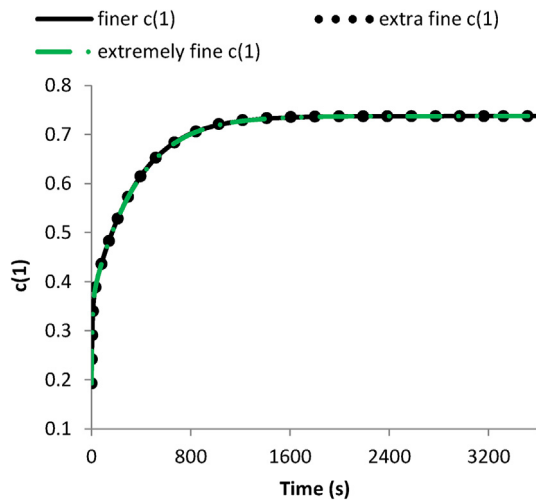


Fig. 3 – Grid independence-fraction of hydrogen absorbed at different grid sizes.

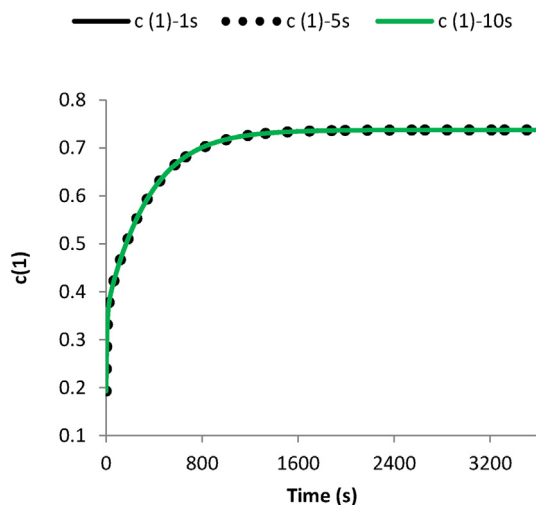


Fig. 4 – Time step independence-fraction of hydrogen absorbed at different time steps.

the system to different design and operating parameters was noted.

Fig. 5 shows the response of the system to change in different properties and operating parameters. It is clear that thermal conductivity and pressure have the most significant impact on the system performance. As the thermal conductivity is increased from 1 to 3 W/m/K, the system performance improves by over 10%. However, when the thermal conductivity is increased beyond 10 W/m/K, the system performance improves at a much slower rate. Similar trend can be observed for supply pressure also. Increasing the supply pressure by 5 bar improves the system performance by around 10%. The system performance is also sensitive to the temperature of the heat transfer fluid, however, it doesn't seem to have the same impact as pressure or thermal conductivity. Variation of heat transfer coefficient from a value of 4.023 W/m²/K (natural convection in air) to 500 W/m²/K, leads to less than 20% improvement in system performance.

Interestingly, the system shows a higher sensitivity to heat transfer coefficient at lower values as can be noted in the transition from 4.023 to 50 W/m²/K. While the system performance is highly sensitive to porosity, any change in porosity from a base value of 0.5 adversely affected the system performance. Therefore, from the above analysis, it was realized that in order to improve system performance, thermal conductivity and pressure were the two most important parameters.

A similar analysis for the system was also carried out for desorption. The reference pressure and HTF temperature were 1 bar and 323 K respectively. The other parameters were varied in the same manner as in case of absorption. Fig. 6 shows the sensitivity of hydrogen desorption to various parameters. Like absorption, the dependence on pressure and thermal conductivity seems to be overshadowing the effects of other parameters. Other than porosity, the increase in which appears to impact hydrogen desorption positively unlike hydrogen absorption, the other parameters such as heat transfer coefficient and HTF temperature do not have a marked influence on hydrogen desorption.

It is also important to point out that system scale can have a significant bearing on the results. For instance, in case of smaller systems where diameter is less than 8 mm, a heat transfer material such as aluminum foam may not be required [3], whereas in case of larger systems as pointed out in the analysis, the same does not hold true. Similarly, heat transfer coefficient and temperature of cooling fluid may have a greater significance in case of smaller diameter systems where heat transfer from the core of the metal hydride bed is transferred to the wall without much resistance [32,33].

Effect of improvement in thermal conductivity

In order to improve the thermal conductivity of the metal hydride bed, metal foam, ENG compacts, or a network of fins can be used. While ENG compacts have exhibited better performance than foam in terms of thermal conductivity however, the deterioration of ENG compacts could be an issue [6]. Between fins and foam, it was observed by Ferekh et al. [10], that the improvement in heat transfer using metal foam is better than with fins. Moreover, foam is more uniformly distributed within the alloy bed as compared to fins, therefore, can contribute to a more uniform heat transfer. Therefore, in the present case aluminum foam will be used for improvement of thermal conductivity. From the sensitivity analysis it was observed that any improvement in thermal conductivity beyond 10 W/m/K was not very effective in improving system performance. Therefore, aluminum foam of porosity 91% which leads to an effective thermal conductivity of 11.45 W/m/K was used. The system with enhanced thermal conductivity was observed to absorb hydrogen much faster, especially under forced convection conditions as can be observed from Fig. 7. It should be noted that system performance here on is discussed in terms of percentage reaction completion (hydrogen absorbed) with respect to time. The heat transfer coefficient has been assumed on the basis of the known values of heat transfer coefficients: natural convection in air 4.023 W/m²/K and around 1000 W/m²/K for forced convection in water.

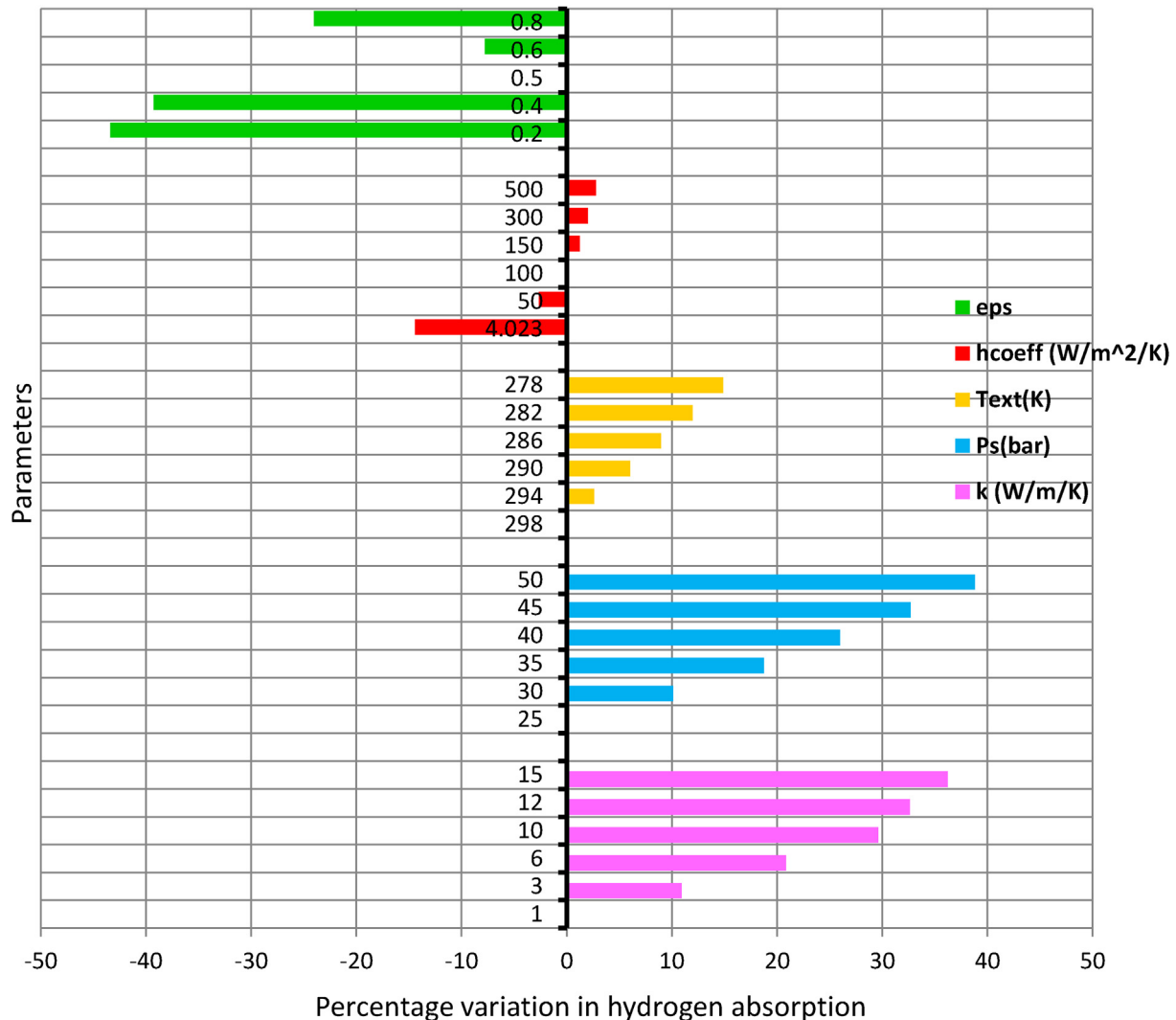


Fig. 5 – Percentage variation in hydrogen absorption with change in thermal conductivity, pressure, HTF temperature, heat transfer coefficient, porosity.

Fig. 8 depicts the evolution of total heat flux streamlines, with the change in colour from blue to red indicating an increase in magnitude of the heat flux. In case of natural convection with no thermal conductivity enhancement, convection currents can be observed from Fig. 8 (a), which rise due to the heat generated by reaction taking place on the alloy surface but for the most part of the bed no significant heat transfer is taking place. At 500 s, however, heat transfer can also be observed in the near wall region. Although the radial heat transfer in this case is far less developed than the enhanced thermal conductivity cases i.e. Fig. 8 (e) and (f), where the streamlines are originating from the core of the bed and the magnitude of flux is also much higher. On comparing Fig. 8 (b) and (c), it can be noted that in case of forced convection the reaction has not only moved closer to the core than natural convection, the magnitude of heat flux is also much larger. Also in the figures at 50 s, strong convection currents can be observed in the expansion volume regions which have weakened at 500 s. As discussed earlier,

it is because in the early stages of the reaction the metal hydride bed surface is a fast reaction zone and the heat thus released is taken up by the gas present in the expansion volume and contributes to the gas motion via thermal buoyancy.

Addition of water jacket

In the previous section it was observed that the system showed much better performance with addition of aluminum foam under forced convection in water. In the next step, a water jacket is added to the system, through which a fixed mass of water flows per unit time, determined by simple energy balance. The width of the water jacket was decided on the basis of ASME pressure vessel code (section VIII) and optimized on the basis of the system performance. Fig. 9 (a) shows a half-sectional view of the reactor with jacket and aluminum foam. The performance of the system is seen to depreciate upon increasing the jacket width as can

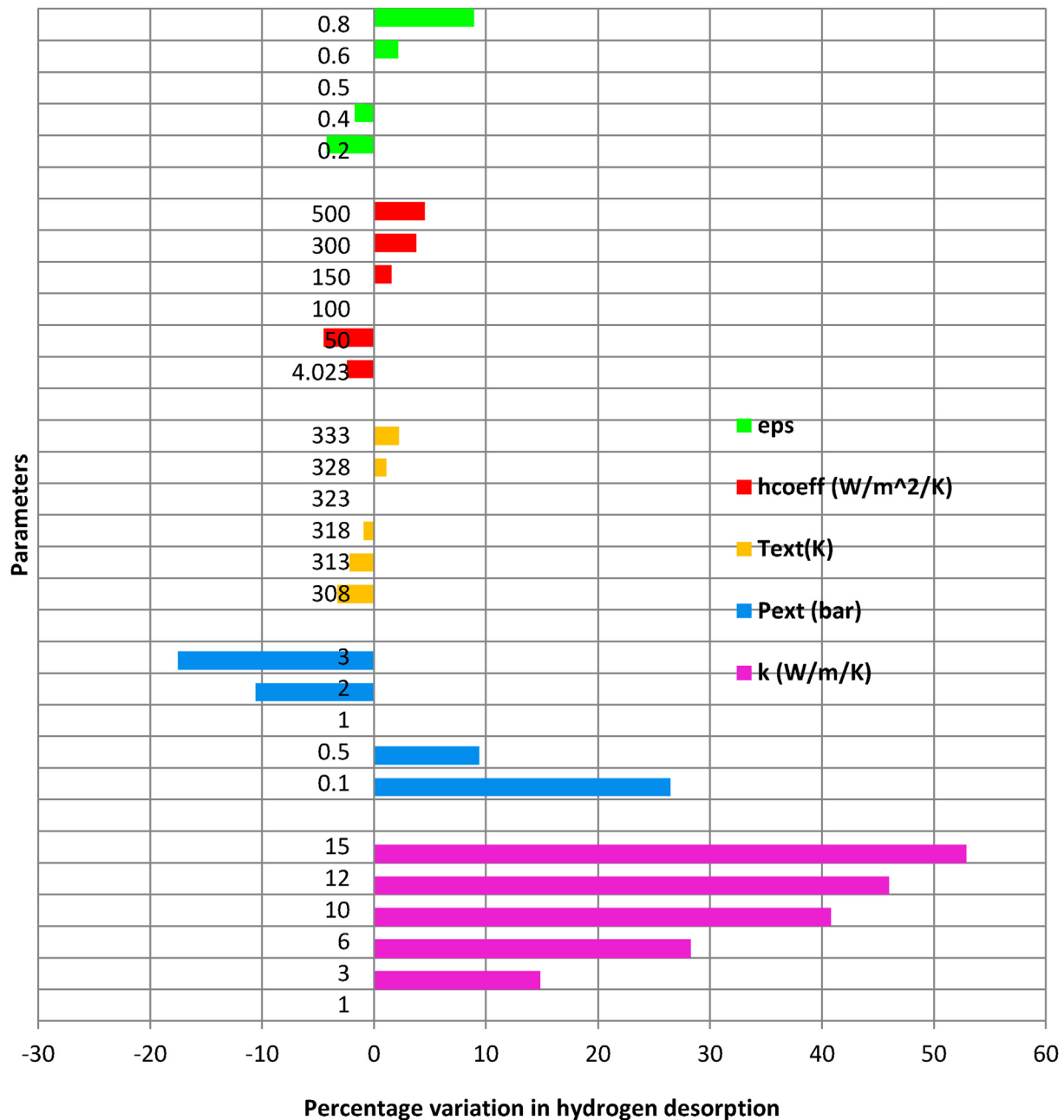


Fig. 6 – Percentage variation in hydrogen desorption with change in thermal conductivity, pressure, HTF temperature, heat transfer coefficient, porosity.

be noted from Fig. 9 (b). This could be attributed to the subsequent decrease in Reynolds's number and hence the heat transfer coefficient (since mass flow rate is kept constant). Therefore, it was noted that 0.0278 m (2.78 cm) was the most feasible jacket width.

As can be seen from Fig. 9 (b), the performance of the system does not meet the requirement of 80%–90% reaction completion even in 3600 s. Therefore, further improvements in design are required to meet that objective. Since heat transfer is known to be a major bottleneck in hydrogen absorption/desorption, consequently we look at the methods

through which heat transfer can be further improved. Firstly, the thermal conductivity can be improved. Secondly, heat transfer area can be added to the system such as fins, cooling tubes. Thirdly, the heat transfer parameters such as heat transfer coefficient, heat transfer fluid temperature etc. can be worked with. Decreasing the heat transfer fluid temperature below the ambient condition will lead to the addition of cooling equipment, adding to auxiliary weight and costs. The first and the third techniques have already been explored to some extent. The addition of heat transfer area in the form of cooling tubes will now be investigated.

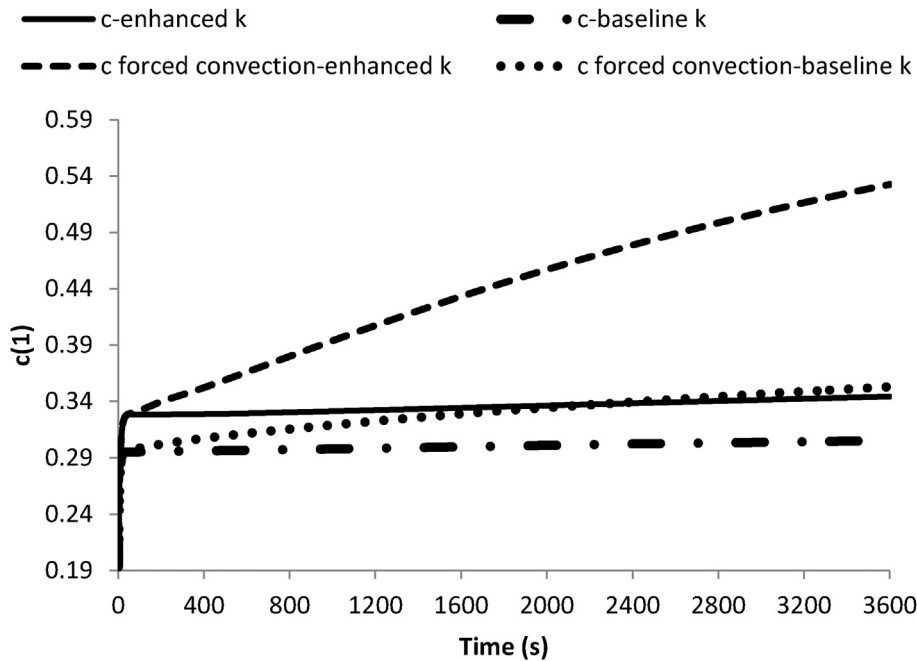


Fig. 7 – Comparison of fraction of hydrogen absorbed with and without thermal conductivity enhancement under free and forced convection conditions.

Addition of cooling tubes

The study was conducted with one, five and nine cooling tubes of three different sizes based upon the standard dimensions. Since the embedded cooling tube model exceeds the conformability limit of 1 m height significantly, therefore, it will have to be divided into two reactors of radius 12.725 cm and length 1.004 m. The jacket width was also changed accordingly. Fig. 10 (a) shows a section of the MH bed with nine cooling tubes, one at the center and two in each quadrant, uniformly spaced. Upon increasing the number of cooling tubes from one to five a small improvement in the hydrogen absorption rate was noted as shown in Fig. 10 (b). However, when the number was increased from five to nine, the subsequent improvement was even lesser. Increasing the diameter of the cooling tube also had a positive impact upon the system performance. The reaction fraction achieved in 3600s even with 9 cooling tubes is only around 66%.

Clearly, thermal conductivity, heat transfer coefficient, and heat transfer area are not the only limiting parameters. The radius of the MH bed is pivotal in ensuring effective heat transfer as was also pointed out by Gopal and Murthy [34,35], Visaria et al. [36]. In order to study the effect of bed radius, the temperature variation across different standard size bed radii was studied. As can be seen from Fig. 11, the larger the bed radius, larger the range over which the temperature is constant. As the bed radius decreases not only does the temperature decrease but the rate at which it reduces also becomes sharper. Since opting for the smallest radius will not be the optimum choice considering system size and cost. Therefore, middle range values of radii which have a reasonably good temperature drop pattern were tested.

Multi-tube metal hydride reactor

As was observed from Fig. 10, increasing the cooling tubes did not improve upon the system performance to the acceptable range of 80–90%. Also Fig. 11 depicted that the temperature within the core of the reactor remained constant for large diameter beds, therefore the bed radius was further reduced. Clearly with a radius around half than that of the original tank, multiple tubes will have to house the amount of metal, if the reactor height is to remain constant. Fixing the height of the tank at 1 m, varying the radius of the tube, the number of tubes required can be found since the volume of metal is known. The process is repeated till a feasible system height is achieved. In the present case, with a radius of 0.073156 m (7.32 cm), the number of cooling tubes was taken as 7, resulting in a metal bed height of 0.7134 m and a total height of 0.8561 m, which is within the conformability limits. The system was designed along the lines of a shell and tube exchanger, with the shell diameter given by Ref. [37]:

$$Ds = \frac{N_t \cdot CL \cdot PR^2 \cdot d_o^2}{0.785 \cdot CTP} = 0.635m \quad (15)$$

where, N_t = no. of tubes = 7, CL = 0.87 represents the incomplete coverage of the tubes by the shell, CTP = 0.93 is the constant for single-pass tubes, PR is the pitch to tube outer diameter ratio, d_o = 0.16828 m (16.828 cm) is the outer diameter of the metal tube. It may be noted that a triangular tube pitch with a PR value of 1.25 was chosen. Triangular tube pitch is more compact and the value 1.25 falls within the recommended range of 1.25–1.5. With the shell diameter known, a 2-D model of the tank was developed as shown in Fig. 12.

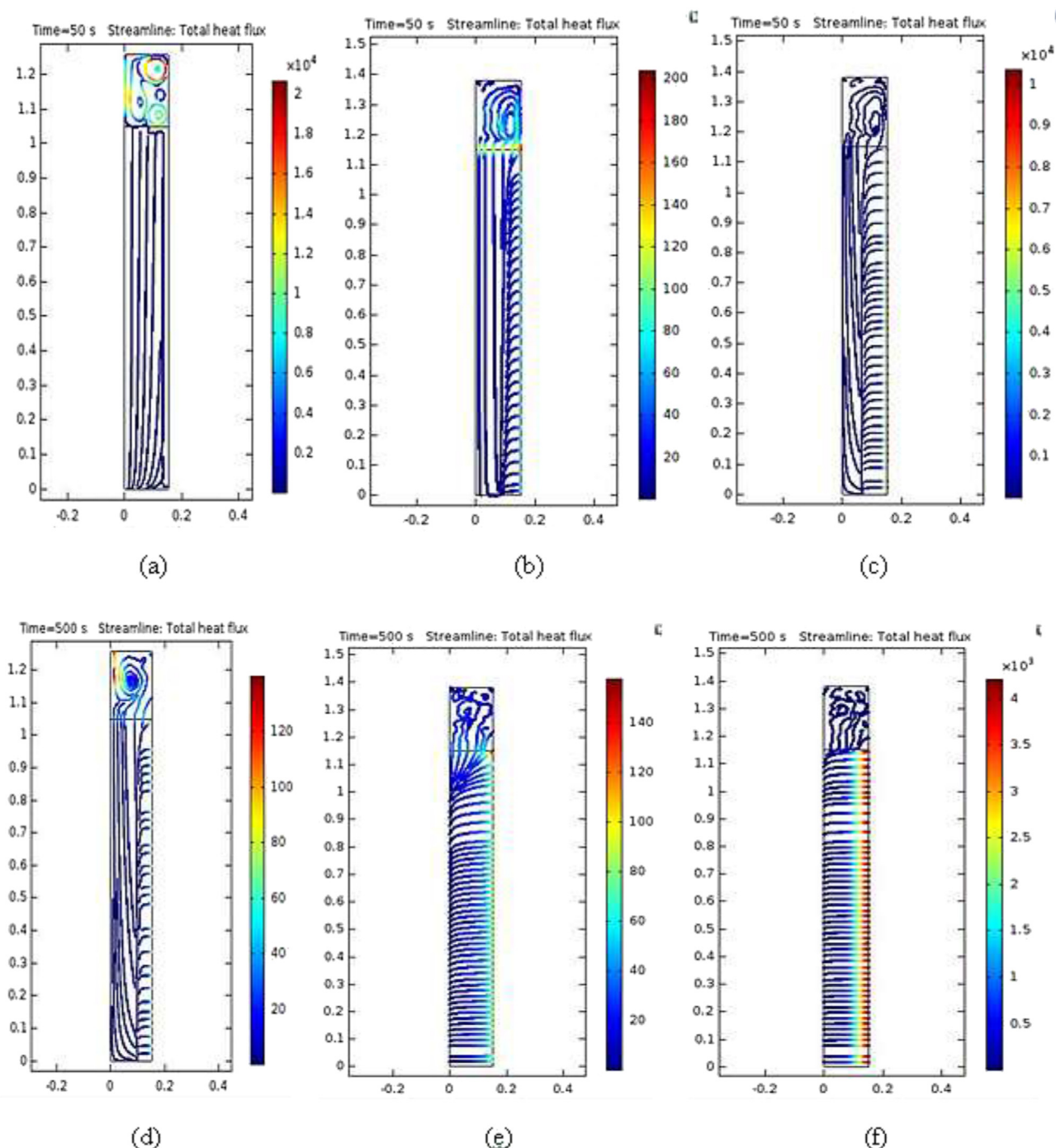


Fig. 8 – Heat flux streamlines at 50 s and 500 s in (a) and (d) natural convection with no k enhancement; (b) and (e) natural convection with aluminum foam; (c) and (f) forced convection with aluminum foam.

The heat transfer coefficient for the aforesaid design was calculated by the Kern's method and was found to be $650.89 \text{ W/m}^2\text{K}$. Upon simulation it was observed that while system performance improved over the single tank scenario, it was almost the same as the system with cooling tubes. Hence, the tube radius was reduced further to 0.0486 m (4.86 cm). The corresponding number of tubes was found to be 14 with a metal bed height of 0.8 m and system height of 1 m . This resulted in a shell diameter D_s of 0.583 m and a heat

transfer coefficient of $850 \text{ W/m}^2\text{K}$. A comparative analysis of the multi-tubular tank with 7 and 14 MH tubes, and the single tank with nine cooling tubes, shown in Fig. 13, was done.

The evolution of heat flux streamlines in the reactors I i.e. with nine cooling tubes, II i.e. with 7 MH tubes and III with 14 MH tubes is compared in Fig. 14. On the colour scale, blue represents lower magnitude of heat flux and red the highest. In Fig. 14 (a) and (b), from the orientation of the heat flux streamlines it can be seen that in the initial phases of the

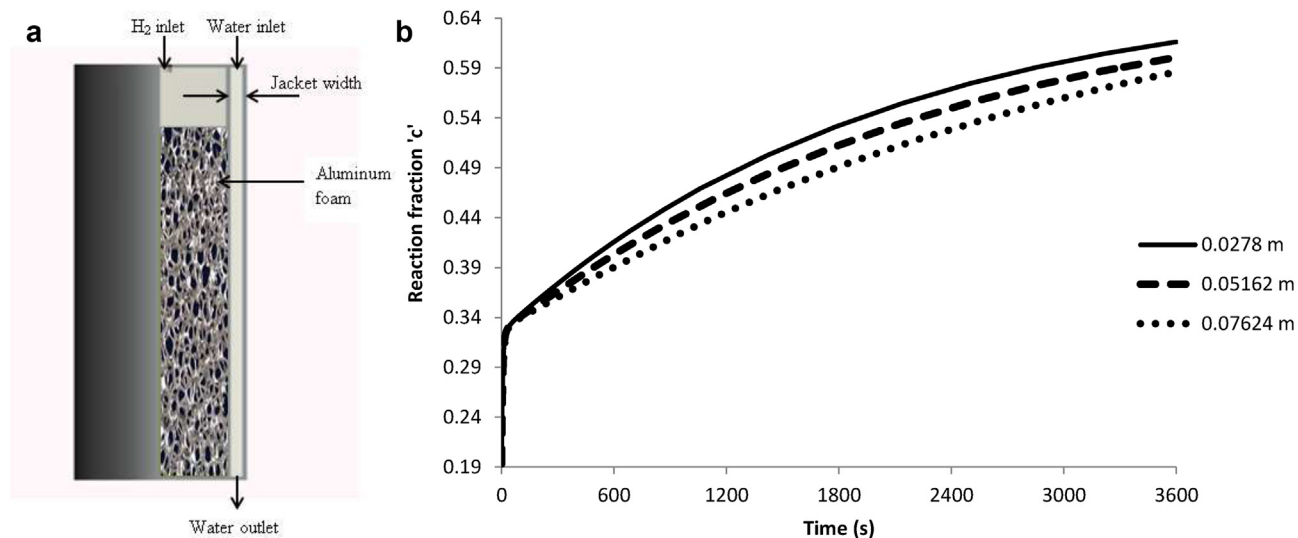


Fig. 9 – (a) Half-sectional representation of the MH reactor with aluminum foam and external water jacket. (b): Comparison of fraction of hydrogen absorbed in MH reactors with aluminium foam at different external water jacket widths.

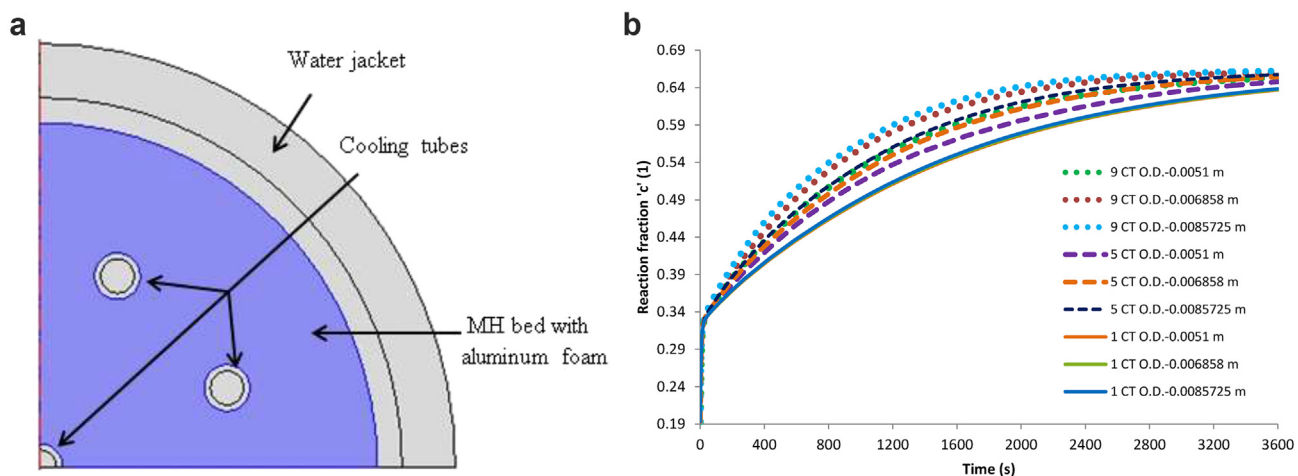


Fig. 10 – (a) MH bed with aluminum foam and nine cooling tubes surrounded by a water jacket. (b): Effect of addition of cooling tubes on fraction of hydrogen absorbed with one, five and nine cooling tubes of three different diameters.

reaction i.e. at 50 s, the external jacket seems to be a more active site of heat transfer as compared to 500 s. Each cooling surface has its zone of influence from which it draws heat. This would lead to the conclusion that more number of cooling surfaces would improve heat transfer leading to a higher reaction fraction; however, due to the fact that the contact area in case of cooling tubes remains small, therefore, the impact may not be as significant. In Fig. 14 (c) and (e), the highest heat flux can be observed near the wall while the core does not show any major activity. At 500 s, in Fig. 14 (d) and (f), while the core registers higher heat flux than before due to more reaction and the subsequent rise in temperature, however, the magnitude is less than that at the wall. This may be due to the higher near wall temperature gradient offered by the cooling fluid.

Fig. 15 shows the comparative performance of the three reactors in terms of temperature and reaction fraction completed. The 14-MH tube multi-tubular reactor exhibits a superior performance as compared to the other two reactors, reaching the maximum reaction fraction in around 1600 s, whereas the 7-MH tube reactor and the nine cooling tube reactor take around 2700 s and 3400 s respectively. Also, observing T_3 and the corresponding reaction fraction c_3 , a very interesting point comes to light. Even though the average temperature is close to room temperature and no heat transfer bottlenecks are present in the system, the reaction rate is still very low. This phenomenon can be explained with the help of the equilibrium pressure Eq. (14). It should be noted here that the modified Van't Hoff equation has been used

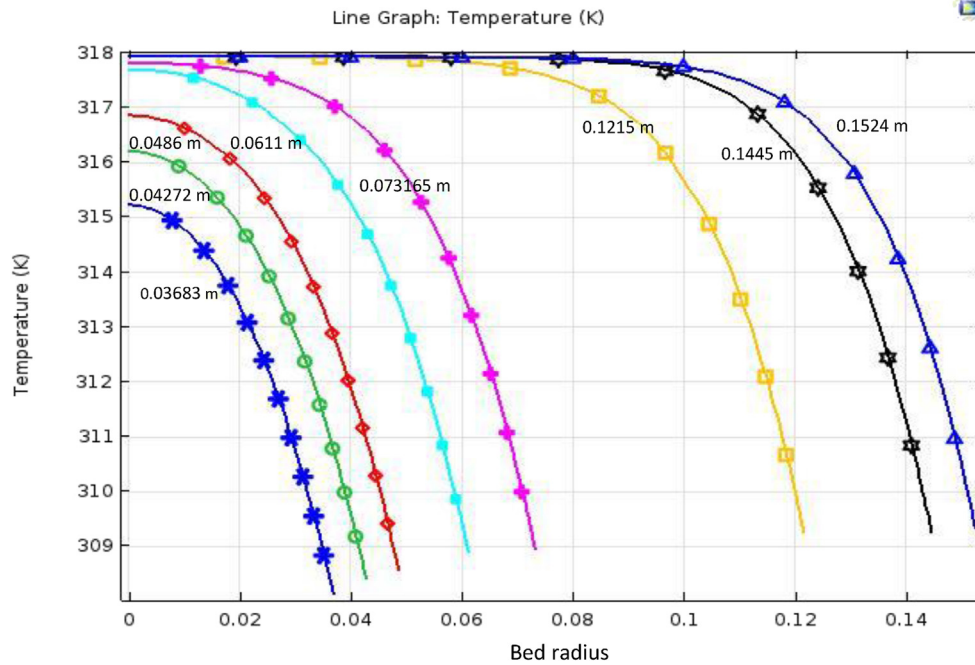


Fig. 11 – Variation of temperature with bed radius for different MH bed radii.

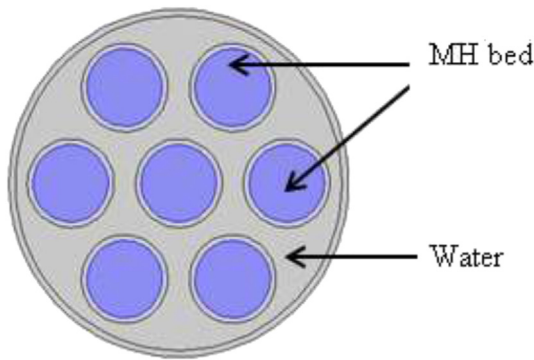


Fig. 12 – Shell and tube type metal hydride storage tank.

which includes the MH concentration ratio term 'c'. The effect of the MH concentration term may not be dropped without loss of accuracy as was noted by Nam et al. [38].

$$\frac{P_{eq}}{P_{ref}} = \left(\exp\left(\frac{S}{R} - \left(\frac{H}{R \cdot T}\right)\right) + (\varphi + \varphi_o) \cdot \tan(\pi \cdot (c - 0.5)) + \frac{\beta}{2} \right)$$

From the above equation, it is clear that once fifty per cent reaction completion is achieved, the term '(c-0.5)' in the above equation becomes positive leading to an addition in the exponential index. This results in an increase in the equilibrium pressure not due to high temperature but as a consequence of a higher hydride concentration in the system. This is quite simple even when seen from an equilibrium reaction perspective.

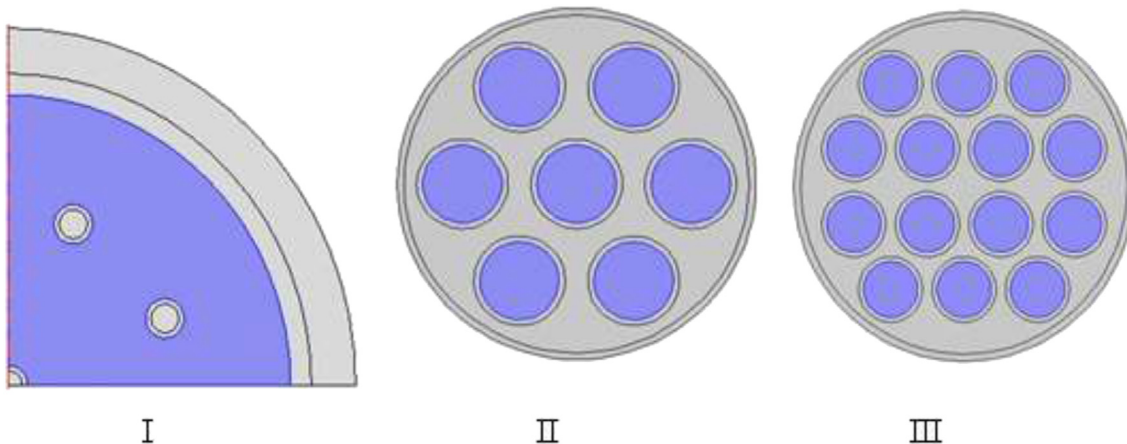


Fig. 13 – Comparison of three different reactor configurations: i) 9 cooling tube system; ii) 7- MH tube shell and tube type system; iii) 14-MH tube shell and tube type system.

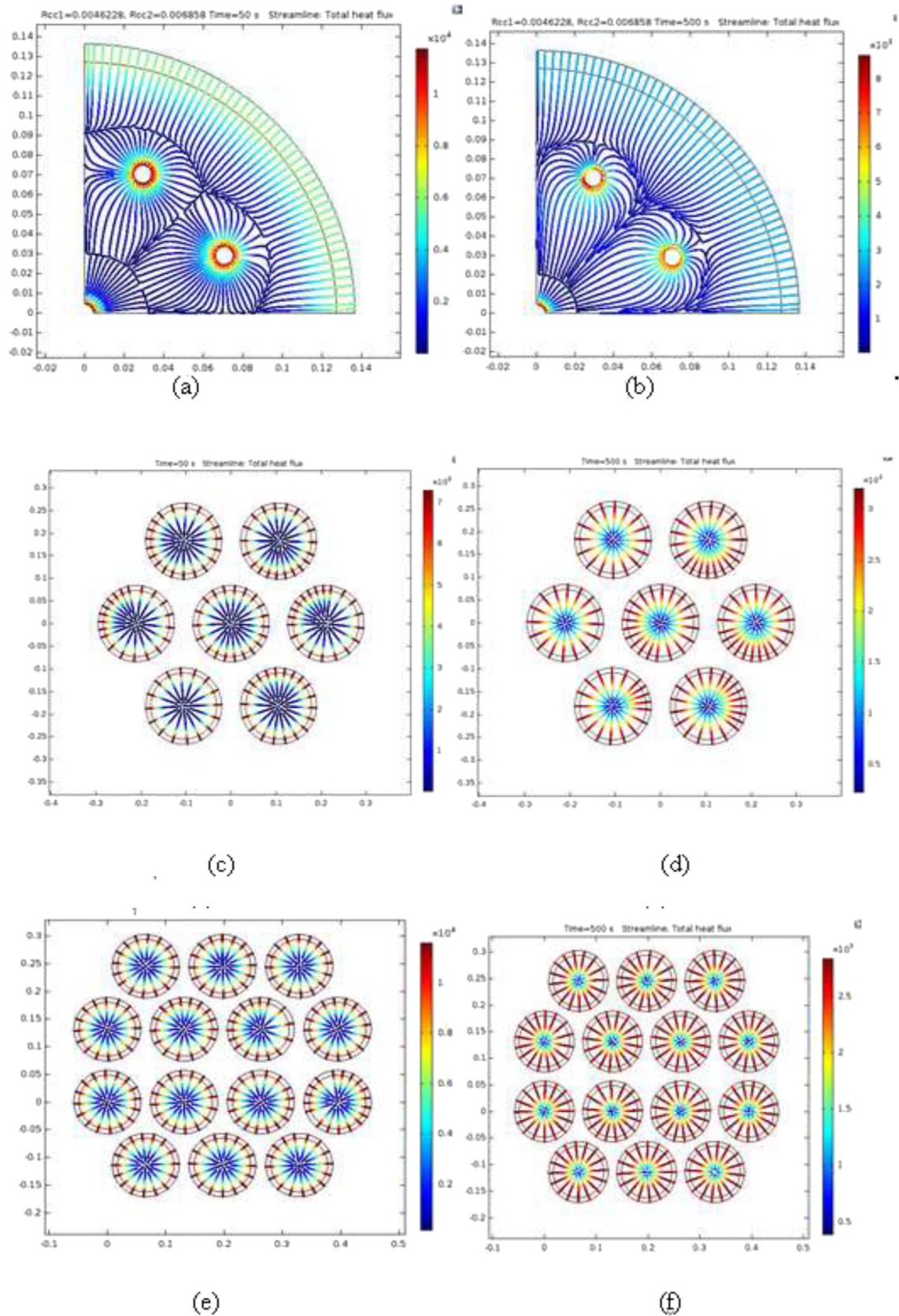
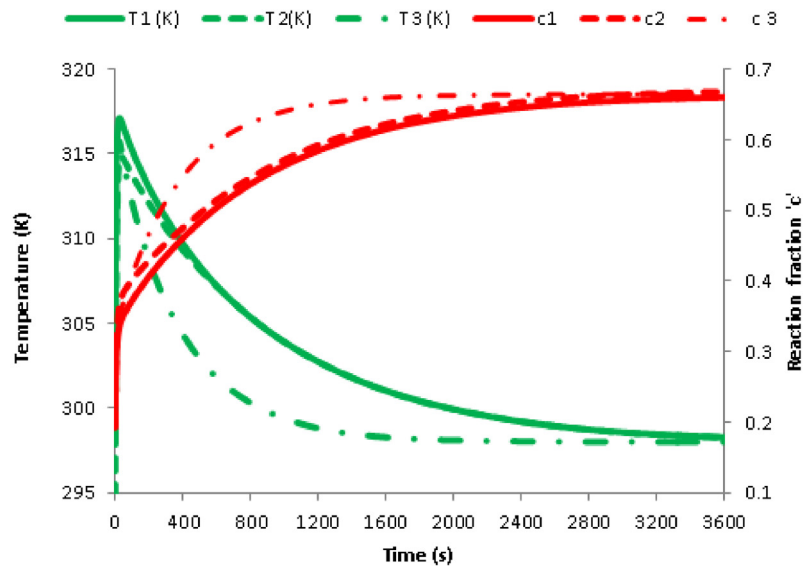
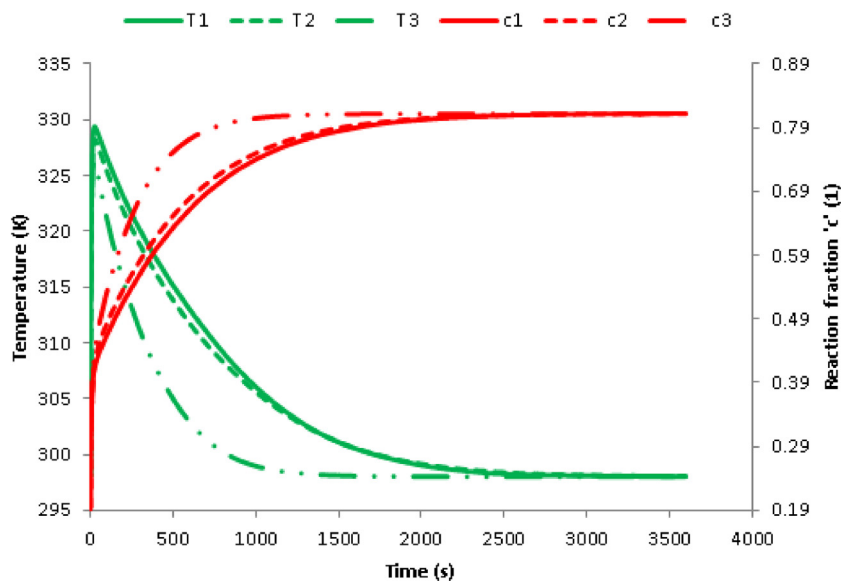


Fig. 14 – Heat flux evolution at 50 s and 500 s: (a), (b) 9 cooling tube configuration; (c), (d) 7 metal hydride tubes; (e), (f) 14 metal hydride tubes.



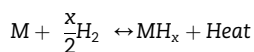
1) 9 cooling tube reactor; 2) 7 MH tubes multi-tubular; 3) 14 MH tubes multi-tubular

Fig. 15 – Comparative performance of the three different reactors at 25 bar supply pressure.



1) 9 cooling tube reactor; 2) 7 MH tubes multi-tubular; 3) 14 MH tubes multi-tubular

Fig. 16 – Comparative system performance of the three different reactors at 40 bar supply pressure.



The above relation also indicates that as the hydride formation increases the reaction will be stalled and the reaction rate will slow down. Therefore, while heat transfer may be an important controlling parameter for a metal hydride formation reaction it is so only during the first half of the reaction. The reaction kinetics may have a stronger impact on the reaction in later phase. This, however, may not be true for all hydrides as the impact of the term '(c-0.5)' would also depend upon the coefficients ϕ and ϕ_0 , which will vary with each hydride.

In order to improve the performance through other means, the system was subjected to an increased supply pressure and its behavior was analyzed. Fig. 16, shows the response of the three systems to an increased supply pressure of 40 bar. The rapid rise in temperature in all three cases is consistent with the absorption of hydrogen, as fraction of hydrogen absorbed 'c' increases, releasing heat of reaction in process. At the supply pressure of 40 bar, a reaction fraction of 80% can be achieved in around 900 s i.e. 15 min in the 14-MH tube system, whereas the same takes around 1800 s and 1900 s in 7-MH tube system and nine cooling tube system respectively. The rapid

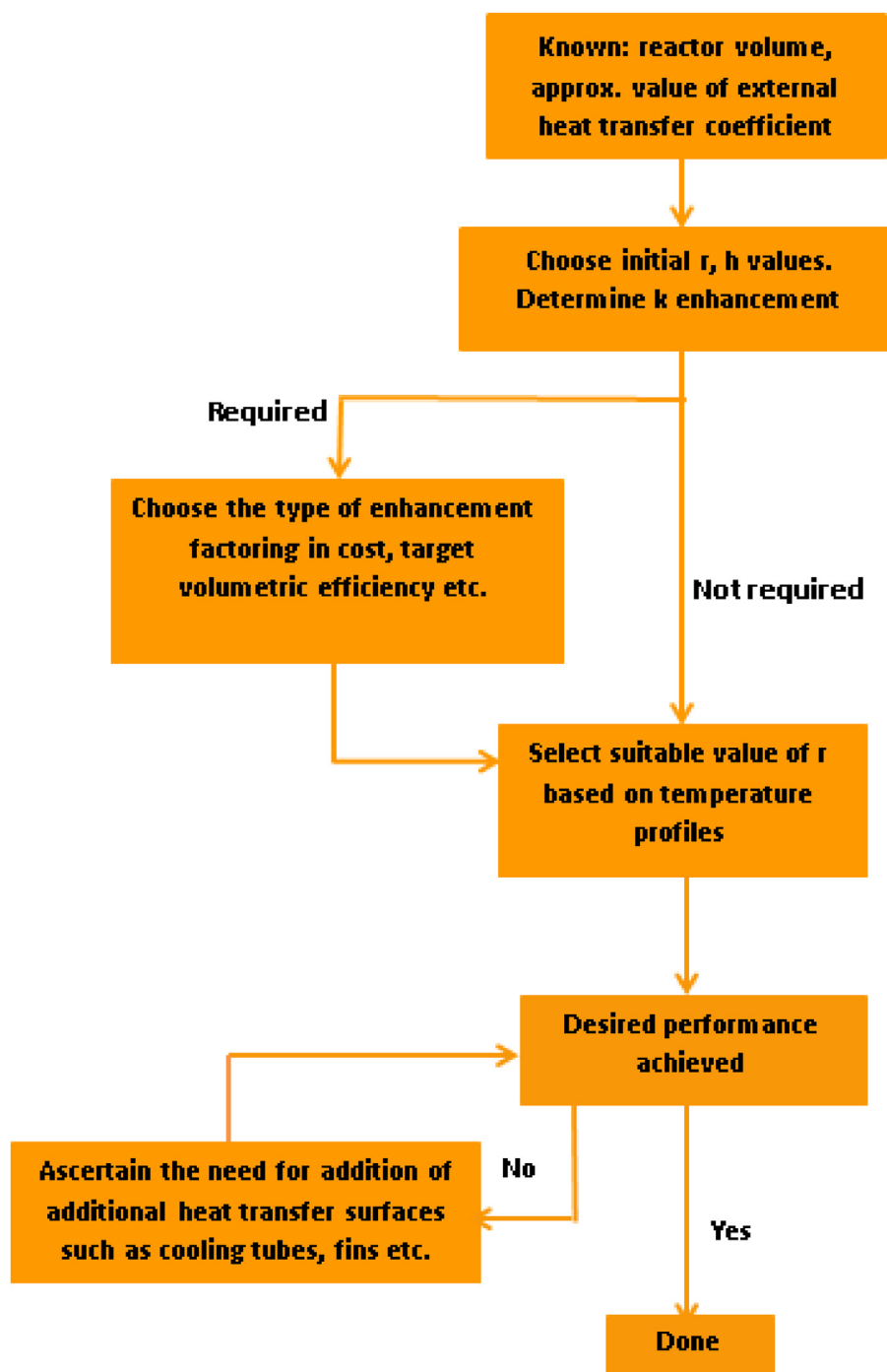


Fig. 17 – Design procedure for a metal hydride hydrogen storage system.

fall of the temperature profiles is indicative of the effective heat transfer in the 14-MH tube system.

Considering, the step-wise process undertaken so far, a design procedure for metal hydride storage system design is suggested in Fig. 17. First the system size is chosen on the basis of the capacity, material etc. If the preliminary system diameter is small (less than 3 cm), then any thermal conductivity enhancements may not be required. If the system is larger, using a thermal conductivity enhancement will

substantially improve the system performance. Once the method of thermal conductivity enhancement has been decided upon, the next step is to check the bed radius which is a critical parameter. While a small bed radius ensures efficient heat transfer, it is difficult to maintain in large systems where system weight could be compromised by using multiple small sized reactors instead of a single large one. There has to be a trade-off between efficient heat transfer and lower system gravimetric capacity, as will be discussed in the next section.

Table 2 – Comparison of empty reactor weights and performances.

Mass	9 Cooling tube (2 reactor system)	7-MH tube	14-MH tube
Mass of shell (kg)	–	45.09161	48.48251836
Mass of tubes (kg)	–	58.8738	77.77248356
Mass of tank (kg)	68.37446	–	–
Mass of ECT (kg)	3.98	–	–
Mass of mesh (kg)	20.4	22.61816	27.9078897
Total mass	92.75446	126.58357	149.5115767
System gravimetric capacity	0.832%	0.749%	0.7%
80% absorption time @ 40 bar (s)	1895	1805	900

After the selection of a suitable bed radius if the system performance is still not satisfactory, additional heat transfer enhancements such as embedded cooling tubes etc. may be included.

System gravimetric capacities

The 14 tube multi-tubular system seems to outperform the other two systems as was seen from Figs. 15 and 16. However, the inclusion of 14 cylinders would also add more weight to the system. Any analysis undertaken for a practical system must weigh the benefits of efficient performance against the additional weight that may be added to the system in such a case. Table 2 presents a comparison of the empty reactor weight of the three systems analyzed in the last section: a) system with 9 cooling tubes b) system with 7 MH tubes c) system with 14 MH tubes. The 14-MH-

tube weighs 1.6 times more than the nine cooling tube system. Also the 14-MH tube reaches 80% reaction completion in less than half the time taken by the latter. Since the objective of the present study is to cater to stationary applications, the 14-MH tube configuration will be the most suitable design, as system weight is not a constraint. Also, in comparison to the large-scale systems previously studied the system performance and gravimetric capacity in the present case is quite promising considering that the alloy used in those cases was sodium alanate which has an inherently higher gravimetric capacity compared to the Ti alloy [20,21,25]. However, the hydrogen absorption capacity of sodium alanate diminishes upon cyclic usage and the kinetics are also not as feasible as those of the Ti–Mn based alloys [39].

Desorption behavior of the multi-tubular configuration (14 tubes)

Any study on the performance of a metal hydride system will be inconclusive until and unless desorption behavior of the system is also taken into consideration. Since the 14-tube shell and tube configuration exhibited good absorption characteristics, desorption study for the same was carried out. The mathematical model used for the same has been previously reported in literature as discussed in section Initial and boundary conditions. Fig. 18 shows the behavior of 14-tube multi-tubular system at a desorption temperature of 313 K and a pressure of 1 bar. At around time 2000 s, desorption process is completed. It may be noted that the temperature and pressure of desorption are very feasible for the operation of a PEM fuel cell.

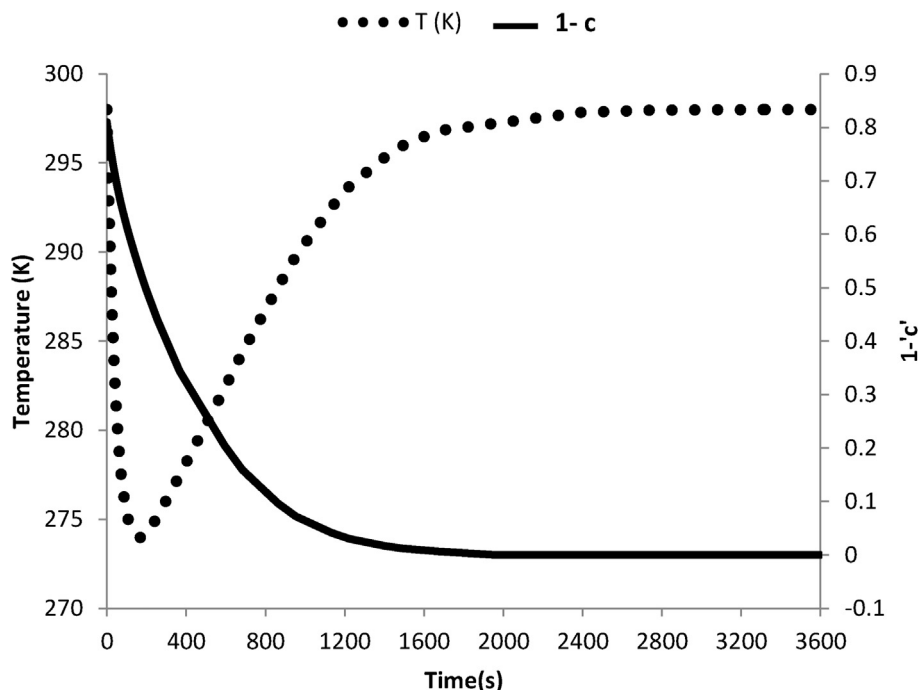


Fig. 18 – Desorption characteristics of 14-tube multi-tube hydrogen storage system.

Conclusions

On the basis of sensitivity to different design and operating parameters such as porosity, thermal conductivity, pressure etc., the performance of a large-scale 210 kg Ti–Mn based MH hydrogen storage system was analyzed. It was noted that the performance was most sensitive to changes in thermal conductivity and pressure. With this as starting point an improved design was suggested. As a result of the series of studies involving different heat exchange configurations, a design procedure was arrived at. Based on the analysis, a 14-MH multi-tube metal hydride reactor was recommended as the best configuration for the present application. At a system gravimetric capacity of 0.7%, the absorption process was completed in 900 s and desorption process in 2000 s demonstrating a superior performance. It was also found that after fifty percent reaction completion, heat transfer ceases to be the major factor controlling system performance as the effect of reaction kinetics becomes more significant.

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